# Customer Growth & Revenue Analysis

How can an online retailer better understand revenue drivers, repeat purchasing and opportunities for data-informed business experiments?

| 541,909 | 530,104 | 19,960 | 4,338 |
|---|---|---|---|
| Source rows | Positive sale lines | Sales invoices | Known customers |

## Analysis workflow

01 Obtain public, historical transaction records from UCI Online Retail.

02 Clean cancellations, nonpositive quantity, and nonpositive unit-price rows.

03 Calculate invoice-level and customer-level measures with Python + SQLite SQL.

04 Export monthly, geography, product and repeat-purchase aggregates.

05 Present results in Excel, a Streamlit dashboard and this decision brief.

## Core metrics and definitions

Positive sales = sum(quantity x unit price) for included line items.

Orders = distinct non-cancelled sales invoice numbers.

Average invoice amount = positive sales / invoices = £534.40.

Repeat purchaser = known customer with at least two distinct sale invoices.

## Important boundaries

Positive-sales value excludes returns rather than netting them: it is NOT net revenue.

No advertising costs or site visits: marketing ROI and web conversion are unavailable.

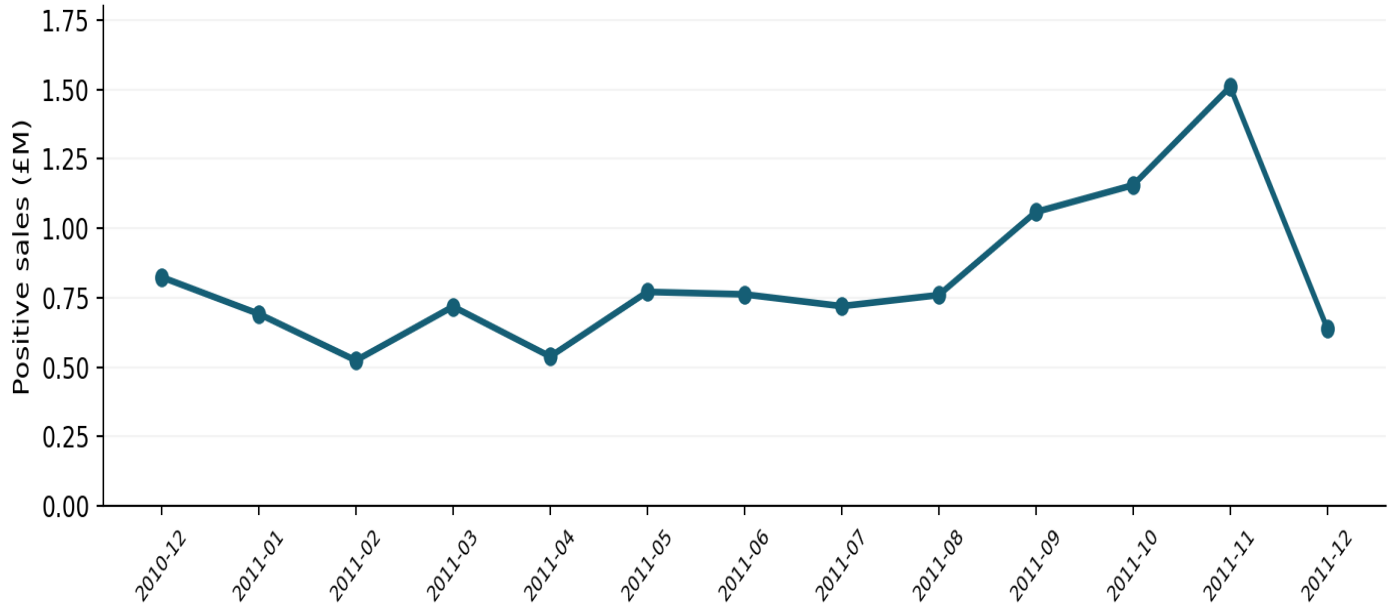
Missing customer identifiers limit customer-level analysis; historic data ≠ current demand.

# What does the data show?

**£10.67M**
Positive sales

**£1.51M**
Nov 2011 sales

**84.6%**
UK share




**MONTHLY POSITIVE SALES**

2011-11 is the highest observed month (£1,509,496). Both December endpoints are incomplete periods; no year-over-year growth or causal claims are made.

## Customer purchasing and data quality

2,845 / 4,338 identified customers (65.6%) purchased on at least two invoices.

132,220 included line items have no customer ID and cannot enter RFM analysis.

DOTCOM POSTAGE is a fee line, not a top merchandise product; audit stock codes.

# From observation to business action

## 01 Test first-to-second purchase reminders

Randomize eligible new customers to reminders or control; allow equal follow-up time.

Measure: 30/60-day reorder rate and incremental sales.

## 02 Evaluate basket-size incentives

Test bundles or shipping thresholds only after obtaining product/shipping costs.

Measure: average invoice amount AND contribution margin.

## 03 Clean catalog and customer identity

Separate charges from merchandise and audit missing customer identifiers.

Measure: product-rank accuracy and customer ID coverage.

## What additional evidence would be required?

Web sessions, ad spend, cost of goods, shipping expenses, and controlled tests.

Current recommendations are hypotheses; no realized business uplift is claimed.

SOURCE: D. Chen, Online Retail, UCI ML Repository. DOI: 10.24432/C5BW33.